

Evaluating Future Paths of Debt in the United States

Jared Bernstein, Daniel Posthumus, Adam Shaw

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1 Introduction

In our [previous paper](#) on the U.S. public fiscal situation, we warned that the “math is looking dangerous”. We showed that conditions formerly favorable to debt sustainability, most notably $r < g$ (the debt’s interest rate being lower than growth), have deteriorated in recent years as real rates rose and growth slowed. We also argued that debt sustainability and the trajectory of the debt had been worsened by Congressional actions that financed historically large tax cuts and spending through deficits. For example, the recently-passed One Big Beautiful Bill Act (OBBBA), according to the [Yale Budget Lab](#), will raise the debt-to-GDP ratio by 52 percentage points from its baseline in 30 years.

One challenge in the debt sustainability literature is that there is no single, widely-accepted metric (by academics and policymakers alike) to distinguish the sustainability of one budget path relative to another. Historically large increases in debt-to-GDP ratios are certainly a signal of rising fiscal vulnerability; however, numerous countries — Japan is an oft-cited example — have experienced such increases in their debt-to-GDP ratio without the kind of acute market stress one might expect.

We therefore introduced a new warning criterion for unsustainability: “ahistorical adjustments” (AA), meaning a situation where external factors — such as a sharp rise in term premia on government debt — necessitate a drastic, sudden adjustment to discretionary fiscal policy, i.e., primary deficit reductions. When that adjustment is “large” in historical terms we argued that the path is potentially unsustainable. Here we take the first step in formalizing and quantifying this definition.

In this paper, we build a stylized stochastic model to more rigorously identify the factors that determine debt sustainability and what constitutes an AA. This model takes external projections of the basic determinants of debt and, through stochastic laws of motion, produces a distribution of fiscal outcomes. The model is deliberately non-structural (i.e., we do not microfound our model); assumptions are both transparent and adjustable to policymakers. Within this model, we are able to simulate and transparently alter assumptions around (1) Congress’s policy responsiveness to the fiscal outlook; (2) the ‘gravity’ of the neutral interest rate r^* ; (3) the persistence of interest rates over time; and — most importantly — (4) alternative assumptions about the mean future path of growth, interest rates, and the primary deficit. In this paper, we present one such example of using the model to evaluate alternative futures, focusing on AI-driven productivity growth counterfactuals. In an accom-

panying policy memo, we present a streamlined presentation of these findings — illustrating our model’s tractability and capacity to contribute to policy discussions.

2 Developing a Model of Debt

To formalize and quantify ahistorical adjustments (AA), we construct a model. The model we build is a variation of a class of probabilistic models known as Stochastic Debt Sustainability Analysis (SDSA), as in [Blanchard \(2024\)](#) and [Favero \(2025\)](#). We use a set of stochastic laws of motion to simulate paths for the relevant determinants of b , the ratio of debt held by the public to GDP: primarily r , g , and s (the interest rate, growth rate, and primary balance, respectively).¹ With this framework, we want to build a model that can be useful to policymakers considering measures that would change the fiscal path. Accordingly we focus on tractability and transparency. Ours is a similar approach to the IMF’s [Sovereign Risk and Debt Sustainability Analysis for Market Access Countries](#) and the European Commission’s [Debt Sustainability Monitor](#).

We develop the system of equations 1-7. Together, they generate a distribution of future debt paths via Monte Carlo simulations. The remainder of this section explains these equations and their logic.

$$b_t = b_{t-1} + \frac{r_t^{av} - g_t}{1 + g_t} b_{t-1} - s_t \quad (1)$$

$$s_t = (1 - c)a_{st} + c \left(\frac{r_t^{av} - g_t}{1 + g_t} \right) b_{t-1} + e_t^s, \quad e_t^s \sim \mathcal{N}(0, \sigma_s^2) \quad (2)$$

$$g_t = a_{gt} + x_{gt} + e_t^g, \quad e_t^g \sim \mathcal{N}(0, \sigma_g^2) \quad (3)$$

$$x_{gt} = x_{g,t-1} + e_t^x, \quad e_t^x \sim \mathcal{N}(0, \sigma_x^2) \quad (4)$$

$$r_t = r^* + \beta_r b_{t-1} + \rho(r_{t-1} - r^* - \beta_r b_{t-2}) + e_t^r, \quad e_t^r \sim \mathcal{N}(0, \sigma_r^2) \quad (5)$$

$$\Delta r_t = \beta_r(\Delta b_{t-1} - \rho \Delta b_{t-2}) + \rho \Delta r_{t-1} + (e_t^r - e_{t-1}^r) \quad (6)$$

$$r_t^{av} = \sigma \cdot r_{t-1}^{av} + (1 - \sigma) \cdot r_t \quad (7)$$

where b_t is the debt-to-GDP (ratio), s_t is the primary balance (surpluses appearing as positive), r_t is the long-term real rate (effectively the real 10-year interest rate), and r_t^{av} is the average effective rate on outstanding debt — all indexed by year t .

Within this framework, we add two concepts. Here we define a debt-stabilizing primary balance, s_t^{stab} and the stabilizing adjustment, AA_t :

$$s_t^{\text{stab}} = \frac{r_t^{av} - g_t}{1 + g_t} b_{t-1} \quad (8)$$

$$AA_t = s_t^{\text{stab}} - s_t \quad (9)$$

¹See Celasun and Ostry ([2006](#)); Cottarelli and Moghadam ([2011](#)); Berti ([2013](#)); Demirel and Otterson ([2022](#)); and Auerbach and Yagan ([2025](#)). Celasun and Ostry (2006), employing a “fan chart” approach, were particularly pioneering for introducing a probabilistic approach to the then-standard debt sustainability framework of “deterministic bound-testing”. The primary motivation for opting for a probabilistic approach is that we don’t rely on a few stylized exogenous shocks to generate a range of outcomes, allowing for greater flexibility and higher fidelity.

Note that, absent fiscal shocks ($e_t^s = 0$), the stabilizing primary balance equals the primary balance in year t if Congress became 100% responsive (i.e., $c = 1$). We interpret AA_t as the one-period primary-balance tightening (in percent of GDP) that would be required in year t to stabilize the debt ratio given realized interest-growth conditions. With the primary balance defined as positive in surplus, $AA_t > 0$ implies a required fiscal tightening, while $AA_t \leq 0$ implies no tightening is required for stabilization in that year.

2.1 Our Approach to Modeling the Interest Rate r

This subsection provides an overview of how we model r ; for more details, see Appendix Subsection 5.2. As shown in Equation 6, we model interest rates' *first difference* to avoid the confounding effects of secular trends in interest rates and debt. Changes in the interest rate depend on the time-invariant real neutral interest rate r^* , lagged changes in debt (scaled by the feedback parameter β_r), persistence of previous interest rates (ρ), and an innovation ($e_t^r - e_{t-1}^r$). We fix $r_0 = 2$ — the approximate average of the 10-year real interest rates post-2022 — and allow Equation 6 to govern its subsequent path. Then, we estimate the average interest rate paid on all outstanding debt, r_t^{av} , as a function of the previous period's average interest rate r_{t-1}^{av} and a 'stickiness' parameter σ that governs how quickly the average borrowing rate adjusts to the current market rate.²

This setup requires the calibration of r^* and the estimation of ρ and β_r . First, though not directly observed, r^* , as estimated by Holston, Laubach, and Williams (2017), has averaged approximately 1% after the Great Financial Crisis, in real terms.³ This is the estimate we adopt for our model and is consistent with the Federal Reserve's most recent Survey of Economic Projections, which has nominal r^* at 3% (1% real plus the Federal Reserve's 2% inflation target).

Second, we have to determine ρ , which governs the persistence of how the interest rate deviates from its neutral value, r^* , and β_r , which governs the magnitude of the pass-through effect of changes in debt on changes in the interest rate. Our estimation strategy for these parameters, relying on first-differencing to avoid our model being overcome by the secular downward trend in the term premium and upwards trend in the debt level, mirrors that of Gust and Skaperdas (2024). We estimate the following equation:

$$\Delta y_t = \beta_r \Delta b_{t-1} - \beta_r \rho \Delta b_{t-2} + \rho \Delta y_{t-1} + \gamma' \Delta \mathbf{X}_t + \epsilon_t, \quad (10)$$

where y_t is the 10-year ACM term premium (Adrian and Moench 2013) — which we use as a proxy for the component of long-term interest rates plausibly affected by debt supply, b_t is the ratio of publicly-held debt to GDP, and $\mathbf{X}_t$ a vector of controls (we keep this lean to avoid overfitting in our model, so that only output gap and the ratio of the primary balance to GDP, s_t , are included).⁴ Then, we set ρ to be equal to the coefficient on Δy_{t-1} .⁵ The full

²We fix $\sigma = 0.8$. In practice, σ will vary in accordance with the maturity profile of outstanding debt. We do not account for such dynamics in our model.

³See Figure 8 for the time-series of the HLW-estimated values of r^* .

⁴The term premium is in percentage points, debt is expressed as a percent of GDP, and all Δ values are quarter-to-quarter changes in percentage points. Thus β_r should be interpreted as the effect of a one percentage point quarterly change in the debt as a percentage of GDP on the term premium.

⁵Specifically, we estimate Equation 10 via Ordinary-Least-Squares (OLS) with Newey-West HAC standard errors.

results from estimating Equation 10 can be found in Table 2. The results, which appear to be robust to splitting the sample around the GFC, show that $\hat{\beta}_r = 0.017$ — meaning a one percentage point quarterly increase in the debt as a percentage of GDP will translate to a 1.7 basis point increase in the term premium — and $\hat{\rho} = 0.158$. This estimate of $\hat{\beta}_r$ falls within the historical estimates for the parameter (for a range of previous estimates of β_r , see Table 1 in the Appendix), toward the lower end of the distribution of prior estimates. This suggests that, in effect, our debt projections should be interpreted as lower bounds; a higher value for $\hat{\beta}_r$ would amplify any modeled debt path. We use these values of $\hat{\rho}$ and $\hat{\beta}_r$ when running our SDSA Monte Carlo simulations. While $\hat{\rho}$ and $\hat{\beta}_r$ are estimated at the quarterly frequency, to be clear, as a calibration, we are applying them to annualized dynamics in simulations.

2.2 Our Approach to Modeling the Primary Balance s

This subsection provides an overview of how we model s ; for more details, see Appendix Subsection 5.3. We model the primary balance as a function of a Congressional policy responsiveness parameter (c), the forecasted primary balance (a_s), an innovation (e_t^s), and the gap between the average interest rate paid on outstanding debt and growth ($r_t^{av} - g_t$). To reflect the fiscal impacts of the recently-passed One Big Beautiful Bill Act (OBBBA), we adopt The Budget Lab’s primary balance projections conditioned on the permanent enactment of the Senate version of the bill as the average of future distributions of the primary balance.⁶

One of the key parameters in the model is c , the Congressional responsiveness parameter. Put simply, this parameter dictates the weight Congress places on stabilizing (i.e., fixing s_t such that $b_t - b_{t-1} \sim 0$, absent innovations) the debt in determining the primary surplus. Auerbach and Yagan (2025) find that Congress has exhibited no policy responsiveness from 2004 to 2024. They find that between 1984 and 2003, Congress reacted to a 1% higher CBO-projected deficit with 0.15pp of realized deficit reduction (both as a % of GDP). On the other hand, from 2004 to 2024, a 1% higher CBO-projected deficit was associated with realized deficit reduction equal to -0.03pp (in practice, a null effect).

This finding suggests that Congress’s current fiscal policymaking regime is entirely inelastic with respect to the debt level ($c = 0$), a view recent qualitative evidence would support. To illustrate the need for fiscal moderation, we run our model under three assumptions about c ’s value. In the first, $c = 0$ and we assume that, as Auerbach and Yagan (2025) found, Congress’s fiscal policymaking is entirely unresponsive to the level of debt for the next 10 years. We refer to this as the ‘baseline’ fiscal regime, since it reflects Congress’s current behavior. Then, as one counterfactual, we model a more ‘responsible’ fiscal regime, which

⁶This approach to modeling debt policy takes a fundamentally different approach from models whose fiscal policy is incorporated into a *reaction function* framework (e.g., Celasun and Ostry (2006)). We believe parameters estimated on a panel country dataset would not bear external validity when applied to the U.S. context, given its singular fiscal situation. In particular, there is little historical evidence of a measurable fiscal feedback process in the United States — perhaps because, unlike the emerging markets often studied by economists at the IMF, for example, the U.S. is not typically observably punished by capital markets for excessive borrowing. Instead, we leverage existing forecasts as the center of a distribution whose second moment is determined by probabilistic shocks.

— effective immediately — returns to the pre-2004 Congressional regime. In this regime, $c = 0.15$, interpretable as roughly a 15% policy response to projected fiscal deterioration. Finally, in the ‘extreme responsibility’ regime, we fix $c = 0.30$ — what we perceive as the upper bound on plausible fiscal responsibility. We find that return to the fiscal posture of 1984-2003 (and even beyond, the adoption of extreme responsibility) would be inadequate to sufficiently achieve debt sustainability and avoid the need for an ahistorical adjustment.

2.3 Our Approach to Modeling the Real Growth Rate g

We model growth as the sum of three parts. First, the conditional mean in year t is exogenously defined by the path a_{gt} . We fix a_{gt} to the Congressional Budget Office’s (CBO) baseline forecast of real GDP growth. Second, the growth in any period can be shifted by a *transitory* shock, e_t^g , randomly drawn from a normal distribution. Third, the path of growth can be shifted by a *persistent* shock, x_{gt} , which moves the growth rate of the economy onto a different path. This results in a range of growth outcomes, including the most favorable (where shocks are consistently positive) and the least favorable (where shocks are consistently negative).

3 Model Results

We run 20,000 simulations throughout. This section presents three ways to understand the model’s output: (1) projected debt levels; (2) interest-growth contribution to debt dynamics; and (3) long-horizon changes in the debt level. First, Figure 1 plots the distribution of debt levels as a share of GDP (with the interquartile range shaded and the median shown as a bold line). We vary c as described above to match ‘irresponsible’ ($c = 0$), ‘responsible’ ($c = 0.15$), and ‘very responsible’ ($c = 0.30$) fiscal policymaking regimes. For context, we plot historical values of debt held by the public since 2005.

However, as discussed above, debt levels alone are not sufficient for determining debt sustainability (or unsustainability); the dynamics of debt accumulation — how quickly debt rises and the sources of this rise — matter for fiscal vulnerability as well. From the law of motion expressed in Equation 1, we isolate one term, $\frac{r_t^{av} - g_t}{1 + g_t} b_{t-1}$, which captures the interest-growth contribution to debt dynamics. Figure 2 plots this “debt snowball term” for the three different fiscal regimes. For context, we also plot the historical values of this snowball term since 2005. Though there have been recent large increases in debt (particularly around the GFC), Figure 2 illustrates how primary deficits were driving this accumulation; interest and growth, were dampening public debt accumulation (this speaks to the $r - g$ dynamic discussed in our previous piece). Regardless of Congressional responsibility, the part of debt accumulation independent of policy appears set to grow at an unprecedented rate.

Based on these results, we conclude that even were Congress to immediately re-discover the fiscal restraint of the 1984-2003 period, debt and the interest-growth contribution to debt accumulation would each continue to rise. Recall our emphasis on ahistorical adjustments. To illustrate this concept using our simulated results, we plot in Figure 3 the distribution of 10-year changes in the debt-to-GDP ratio (in percentage points) for 1) historical data (1970 - present) and simulated outcomes under $c \in \{0, 0.15, 0.30\}$.

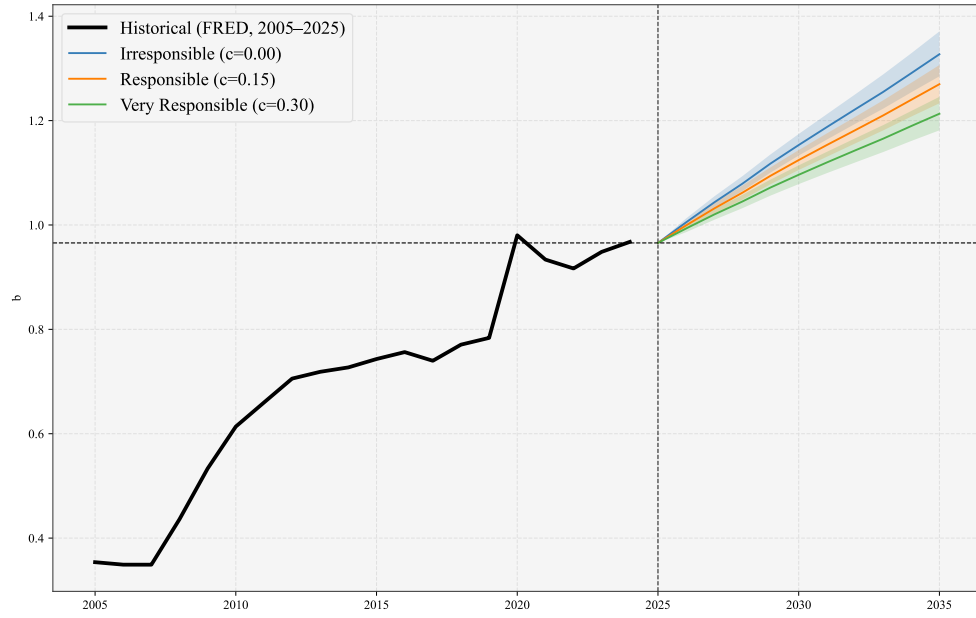


Figure 1: Debt level projections (share of GDP)

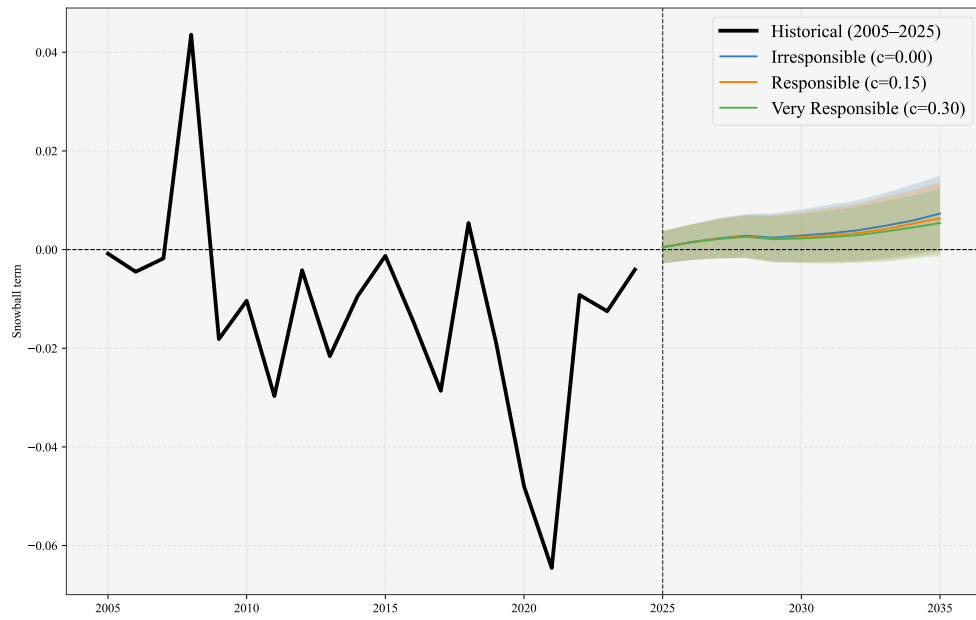


Figure 2: Interest-growth contribution to debt accumulation

These simulated distributions show that the medians of the $c = 0$ and $c = 0.15$ cases are far to the right of the historical maximum. The highly responsible regime ($c = 0.30$) does better, with its lower tail closer to the historical median. What this says in practical terms is that to avoid an AA in the event of a debt crisis, we would require (1) Congress to be far more responsible (in fact, roughly twice as responsive as it was during 1984-2003) and (2) luck (i.e., consistently get positive shocks to growth and rates). Our simulations thus suggest we are *likely* heading for uncharted waters.

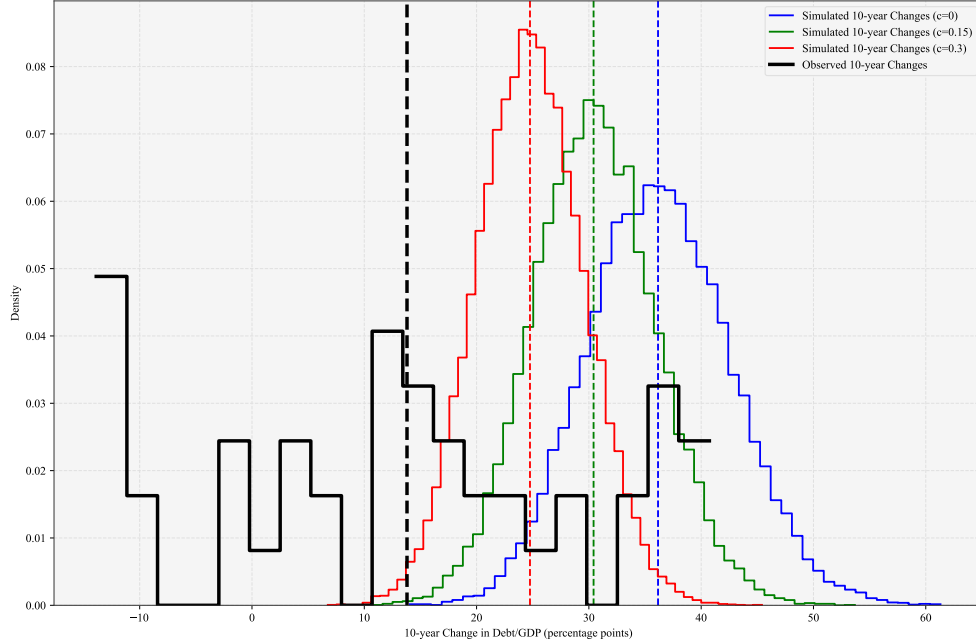


Figure 3: Distributions of 10-year changes in debt, historical and projected

To operationalize our AA concept, Figure 4 plots the historical primary balance (1984-2024) along with the model-implied median primary balance s_t and the corresponding debt-stabilizing primary balance s_t^{stab} under each c regime. Then, the gap $AA_t = s_t^{\text{stab}} - s_t$ (shown in the pop-outs) is the model-implied primary-balance tightening required to stabilize the debt ratio given the realized (1) interest-growth environment and (2) level of debt held by the public. Even under our upper bound fiscal responsibility assumption, the model implies a median required primary balance tightening of 2.4 percentage points of GDP at the 10-year horizon. For context, consolidations of this magnitude — even when phased in over several years — have not occurred in modern history (post-1984) outside of the dot com bubble and post-GFC fiscal consolidation. Under these circumstances, stabilizing the debt — even in the median projection, barring negative economic shocks — would likely require an ahistorical fiscal adjustment.

3.1 Modeling an Artificial Intelligence Boom

Some economists have suggested that AI technology, once fully integrated into the workforce and production, will lead to a sharp increase in productivity growth, and that this will raise

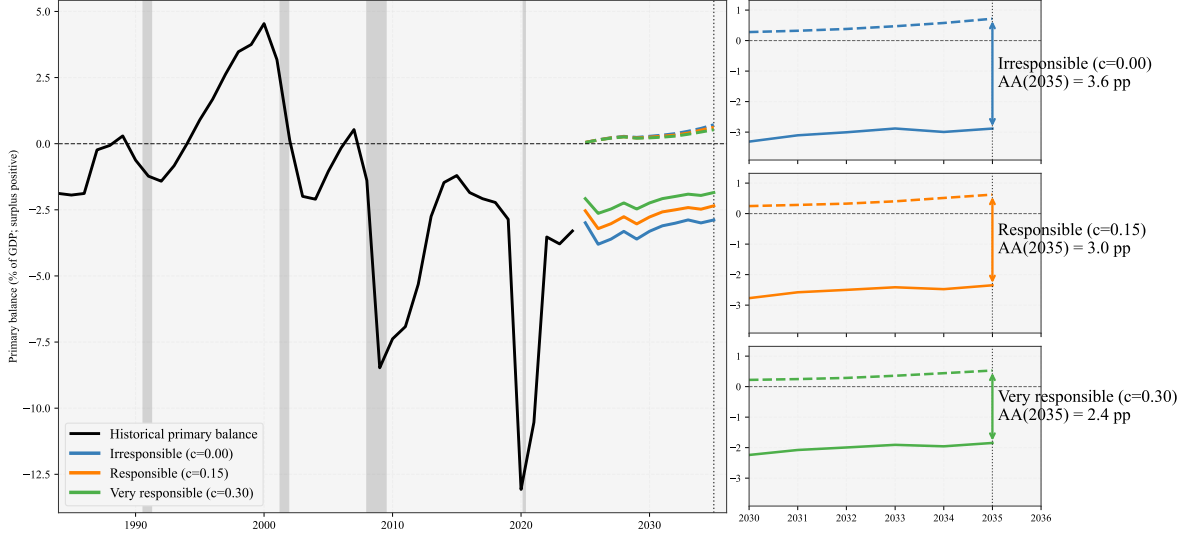


Figure 4: Debt-stabilizing adjustments by level of Congressional responsiveness

g relative to r , thereby staving off the unsustainable scenarios shown above. For example, Torsten Sløk, chief economist of Apollo, released [an edition](#) of his “Daily Spark” newsletter in July 2025 titled “AI Could Solve the US Fiscal Problem”, based on CBO projections. For a sense of the breadth of projections of AI’s impacts on growth, Tom Cunningham has [gathered](#) the most prominent such projections. The Penn Wharton Budget Model (PWBW) [estimated](#) (in a very preliminary analysis whose methodology has not been made public yet) that, through productivity and accordingly GDP gains, AI could reduce deficits by \$400 billion (1.4% of 2024 GDP) between 2026 and 2035.⁷

An attribute of our model is that we can simulate this scenario. In this case, we plug in [CBO’s projected growth rates under an AI boom](#), and again, generate a distribution of debt paths.⁸ Specifically, we use CBO to model the improvement to baseline primary deficits under different assumptions about improved productivity; then we apply this improvement to the Budget Lab’s post-OBBBA primary deficit projections and use this new primary balance as the model input a_{st} and the CBO-projected new growth rate as the model input a_{gt} . We do that here, holding $c = 0$ to match the post-2004 baseline of negligible fiscal responsiveness and modeling three different growth scenarios: (1) CBO’s baseline growth projections, (2) an AI-induced 0.5pp boost to productivity, and (3) an AI-induced 1.0pp boost to productivity. Figure 5 plots the interest-growth contribution to debt accumulation for these three growth scenarios and Figure 6 plots the distribution of 10-year changes in debt against the historical distribution of such changes.⁹

⁷Specifically, they find that AI will increase productivity and GDP by 1.5% by 2035, 3.5% by 2055, and 3.7% by 2075

⁸See Figure 9 for a plotted comparison of CBO’s projected growth rates under an AI productivity boom and the baseline.

⁹We note that the Penn Wharton Budget Model forecasts AI boosts to productivity would also lower inflation; this would, in practice, also lower interest rates as the Federal Reserve would be less likely to raise the federal funds rate without elevated inflation to fight. A lower interest rate would make the debt path lower, suggesting that our estimates in the faster-than-baseline TFP growth scenario should be viewed as

Our model’s findings suggest that a 1.0pp boost to productivity will, even in the absence of any improved Congressional responsiveness, keep interest-growth contributions to debt accumulation negative, such that interest-growth dynamics are working to offset primary deficits. A 0.5pp boost, on the other hand, would stabilize interest-growth contributions to the debt. Simultaneously, such productivity boost shifts the distribution of possible 10-year changes to the historical distribution’s center. Anything short of such an optimistic projection (Cunningham notes that most economists forecast a 0.1-0.5% boost to growth per year), does not appear to ensure fiscal stabilization. However, the gap in the fiscal outcome of the baseline vs. AI boom scenarios is also true to CBO’s growth projections; the gap in our projected fiscal outcomes starts to widen as we approach 2035 (year 10 in the model), which is our operational horizon. The gap widens over time because level effects compound; consequently, the difference between baseline and higher-productivity amplifies over longer horizon. This suggests that were we to extend our model outwards, we would likely see significant improvements to the fiscal situation under AI boom conditions as the horizon lengthens.

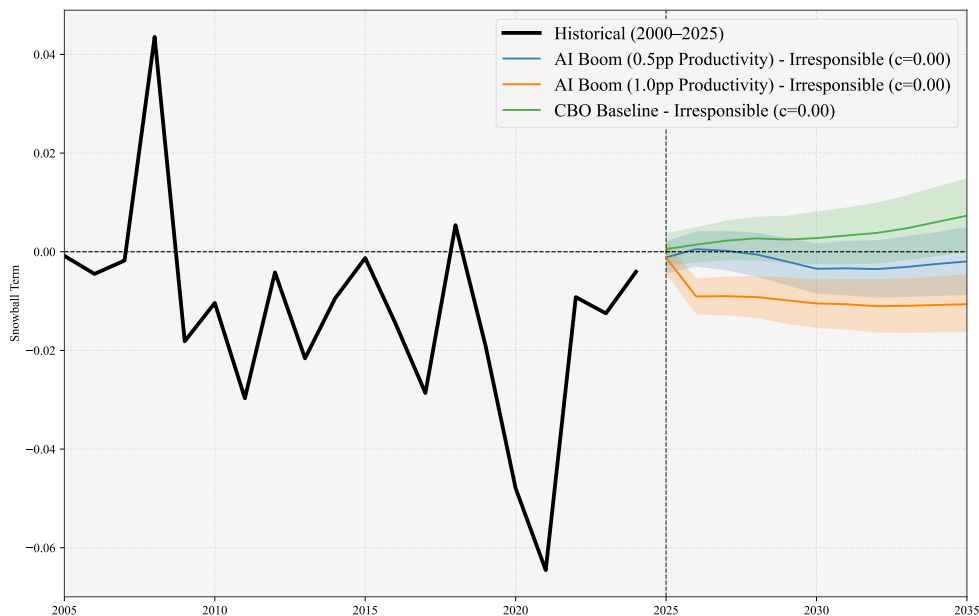


Figure 5: Interest-growth contribution to debt accumulation

Lastly, Figure 7 applies our AA framework to these three scenarios. With a 1.0pp boost to productivity, the model implies $AA_t \leq 0$ in the median projection — i.e., debt stabilization does not require any primary balance tightening. With a 0.5pp boost, the median required primary balance tightening is 1.4 percentage points of GDP. Consolidations of this order are not unprecedented when phased in over several years, having occurred outside of the highly irregular periods of the dot com bubble and post-GFC consolidation. However, our concern in this scenario is concentrated in the upper tail of outcomes, where negative economic shocks could still imply historically large required adjustments.

close to upper bounds for the debt-to-GDP ratio.

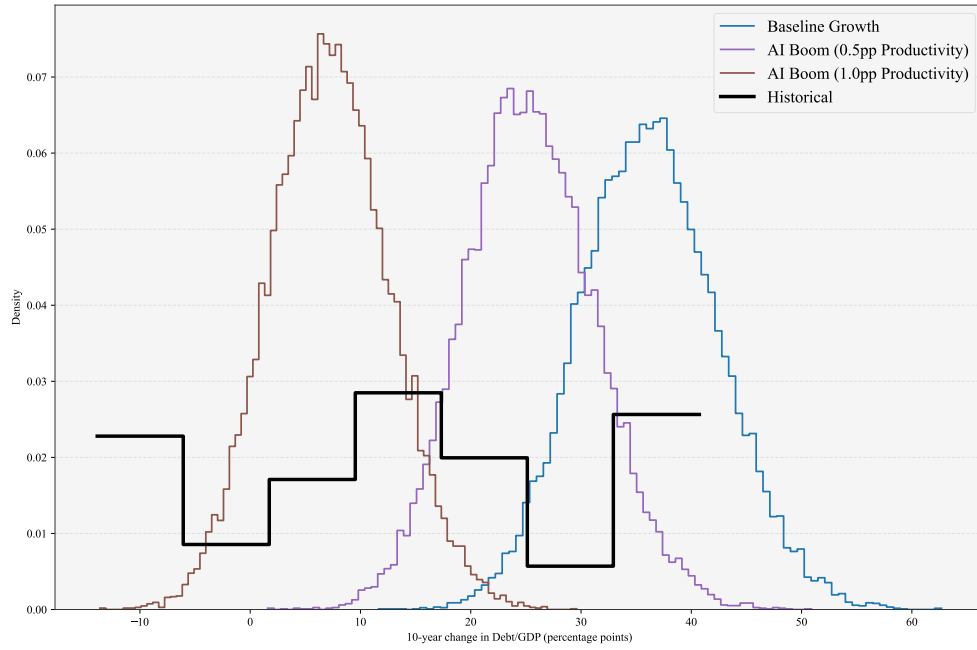


Figure 6: Distributions of 10-year changes in debt as a share of GDP (pp)

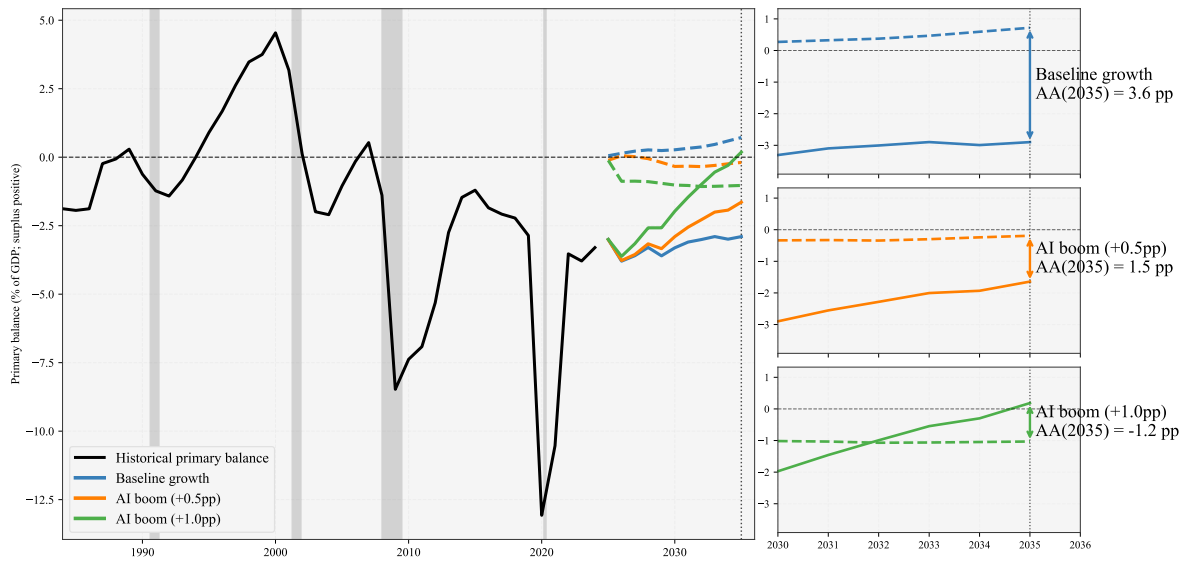


Figure 7: Debt-stabilizing primary balance adjustments by AI growth scenario

4 Strengths and Weaknesses of our Model

Any model involves tradeoffs and the simplification of reality. In our model, which we view as a starting point, we have opted to emphasize first principles and model straightforward, easily-interpretable laws of motion for r , g , s , and b . This straightforwardness is a strength; we believe that this model is more easily digestible to policymakers than other complex (and also deeply helpful) macroeconomic models and, given that the goal of this exercise is to quantify debt sustainability in a way helpful for policymakers with direct control over our fiscal situation, we consider this a strength — even if this simplification comes, in part, at the expense of model richness.

Additionally, the model gives us full control over the mean and distribution of the future path of our key macroeconomic variables r , s , and g . This means that we can use as the first moment of any of these variables projections from external sources; if a bill is proposed, and sources such as the CBO or Yale Budget Lab score these proposed policies on their fiscal/growth/interest rate effects, we can very easily construct counterfactual debt paths that allow us to compare across differing policy baselines. Importantly, this also allows us to model what would happen if, as an example, with the election of a new President in 2028, c went from 0 to 0.15 (i.e., Congress continues its fiscal recklessness until a new administration comes to power, and at that point it returns to its 1984-2003 approach to fiscal matters).

One weakness of our model, as a result of its simplification, is our reliance on linear, independent modeling of r , g , and s (although we do allow for some feedback between r and s). We know that the economy is deeply interconnected; changes in b will result in crowding-out which will not only raise r but also, as a result of higher borrowing costs, lower g . Our model accounts (in a simplistic way) for this first step in crowding-out; however, we would have to enrich our current framework to break out interest rate effects separately into 1) effects on the policy / federal funds rate and 2) private borrowing rates, each of which would presumably affect growth differently. Thus, while the (infamous) findings of Reinhart and Rogoff have been debunked, there is no consensus on the relationship between b and g — meaning it is unclear whether by excluding any causal effect of b on g we risk over- or under-estimating the severity of debt unsustainability (Rahman, Ismail, and Ridzuan 2019).

Furthermore, our model does not take a stance on whether the relationship between debt and interest rates (represented by the coefficient β_r) will fundamentally change in the future; this remains open speculation as we continue to hurdle into uncharted fiscal waters, but it is certainly possible that in instances of a debt spiral, the relationship between b and r is not linear but instead exponential.

One final weakness is that our model does not include the granularity to model changes in the composition of outstanding debt. For example, we make no distinction between the government issuing long-term Treasuries or short-term Treasuries to fund itself; instead, we model a fixed and time-invariant parameter σ which captures the relative strength of the past average interest rate (intended to reflect the average interest rate across all outstanding debt) in determining the future average interest rate. This lack of enrichment prevents our *accurate* projection of debt servicing as a share of GDP (hence our reliance on the curvature of the debt path, which we believe captures in effect the same phenomenon).

What we would like to emphasize is that none of these weaknesses are fundamental to our model; instead, they require enrichments to the template we’ve laid down and we

encourage alterations to this framework to incorporate more non-linear, interrelated, and complex relationships to further improve this model over time.

5 Appendix

5.1 Previous Estimates of Debt’s Effect on Interest Rates

Source	Estimation Period	Effect of 1 pct. pt. increase in debt/GDP (basis points)	Methodology	Term Premium or Expected Short-Rate
Engen and Hubbard (2004)	1976–2003	3 bps	OLS and VAR	both
Gale and Orszag (2004)	1976–2004	4–6 bps	OLS	both
Kinoshita (2006)	1971–2004, international data	4–5 bps	Panel regression	both
Laubach (2009)	1976–2006	3–4 bps	OLS	both
Chadha et al. (2013)	1986–2008	2 bps	OLS	both
Gamber and Seliski (2019)	1976–2017	2–3 bps	OLS	both
Coenen et al. (2012)	N/A	1 bp	Calibrated structural model	Expected short-rate (r^*)
Mian et al. (2022)	N/A	1–2 bps	Calibrated structural model	Expected short-rate (r^*)
Li and Wei (2013)*	1994–2007	6 bps	Estimated term structure model	Term premium

Table 1: Estimates of the effect of a 1 percentage point increase in debt-to-GDP on interest rates (β_r)

5.2 Detailed Explanation of Modeling r

First, here we have plotted the current model r^* estimates from the [Holston-Laubach-Williams \(2023\)](#) model:

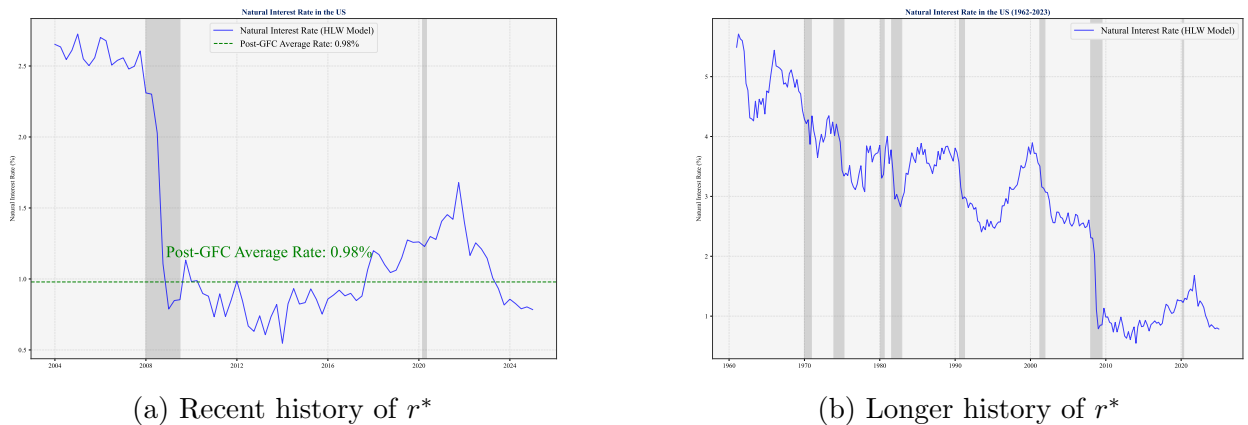


Figure 8: Holston-Laubach Williams (2023) estimates of r^*

Second, to illustrate how ρ works, consider three simple examples: $\rho = 0$, $\rho = \frac{1}{2}$, and

	Full sample	Post-GFC	Pre-GFC
const	-0.008 (0.021)	-0.038 (0.037)	-0.005 (0.025)
d.b.lag (β_r)	0.017* (0.009)	0.013 (0.010)	0.016 (0.047)
d.b.lag2	-0.001 (0.007)	0.001 (0.006)	0.001 (0.038)
d.gap	-0.017 (0.021)	0.018 (0.015)	-0.031 (0.034)
d.primary	0.015 (0.011)	0.019 (0.012)	0.018 (0.039)
d.r.lag (ρ)	0.158** (0.071)	0.151 (0.121)	0.152* (0.085)
R-squared	0.032	0.043	0.030
N	219	66	148

Table 2: ACM 10y term premium: First differences (HAC SEs)

$\rho = 1$:

$$\rho = 0 \rightarrow r_t = \bar{r} + \beta_r b_{t-1} + e_t^r \quad (11)$$

$$\rho = \frac{1}{2} \rightarrow r_t = \frac{1}{2}\bar{r} + \frac{1}{2}r_{t-1} + \beta_r(b_{t-1} - \frac{1}{2}b_{t-2}) + e_t^r \quad (12)$$

$$\rho = 1 \rightarrow r_t = r_{t-1} + \beta_r(b_{t-1} - b_{t-2}) + e_t^r \quad (13)$$

Lastly, below we've included the regression results from estimating Equation 10, under three sample specifications: 1) entire sample, 2) pre-GFC, and 3) post-GFC:

5.3 Detailed Explanation of Modeling s

To illustrate how c works in our model, we can evaluate the two extreme cases, either where $c = 0$ or where $c = 1$:

$$c = 0 \rightarrow s_t = (1 - c)a_{st} + c \left(\frac{r_t^{av} - g_t}{1 + g_t} \right) b_{t-1} + e_t^s = a_{st} + e_t^s \quad (14)$$

$$c = 1 \rightarrow s_t = (1 - c)a_{st} + c \left(\frac{r_t^{av} - g_t}{1 + g_t} \right) b_{t-1} + e_t^s = \left(\frac{r_t^{av} - g_t}{1 + g_t} \right) b_{t-1} + e_t^s \quad (15)$$

We ultimately adopt three values for c : 0 (total irresponsibility, matching findings by Auerbach and Yagan (2025)), 0.15 (a return to pre-2003 fiscal restraint), and 3) 0.30.

5.4 CBO-Modeled Growth Gains from AI Productivity Boom

Figure 9 compares the CBO-modeled AI productivity boom-induced growth rate vs. the baseline projected growth rate.

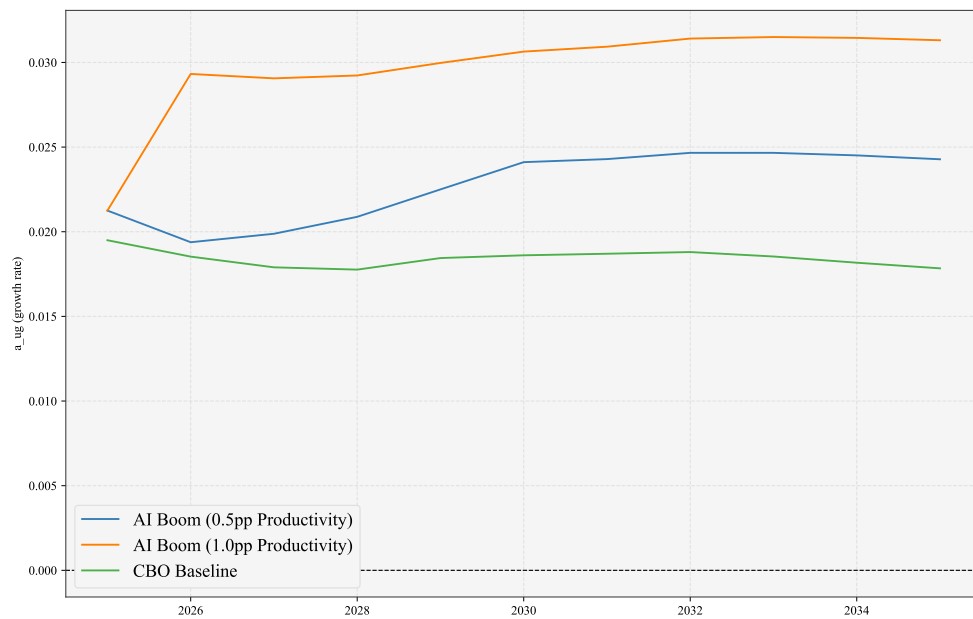


Figure 9: CBO AI boom and baseline growth rate projections

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